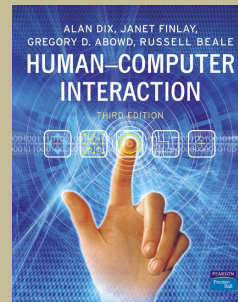


Evaluation

Chapter 9

1

Different chapters for the same topic



Chapter 9



Chapter 8

2

Evaluation

Evaluates both design and implementation
Should be considered at all stages in the design life
Occurs in laboratory / field
Two main categories:
- by experts
- in collaboration with users
Goals:
assess extent of system functionality
assess effect of interface on user
identify specific problems

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Evaluation by experts

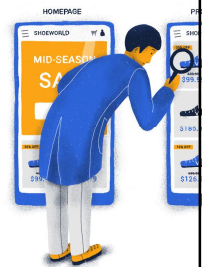
Should occur **during the design process**

If the design itself is evaluated, expensive mistakes can be avoided

Experts **identify any areas that are likely to cause difficulties** because they violate known cognitive principles

These methods are **relatively cheap** since they do not require user involvement.

Limit: they do not assess actual use of the system



4

Expert analysis: main approaches



5

Cognitive Walkthrough (for Learnability eval.)

Evaluates design on how well it supports user in **learning tasks**

Expert **'walks through'** the design to identify potential problems using psychological principles

For each task, walkthrough considers:

- what impact will interaction have on user?
- what **cognitive processes** are required?
- what **learning problems** may occur?



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Cognitive Walkthrough: What do you need

4 things needed (for each task):

1. a detailed specification of the **system** (or a prototype)
2. a description of the (representative) **task** to perform
3. a complete list of **actions** needed to complete the task (a walkthrough scenario)
4. info about **users**



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Cognitive Walkthrough: What do you need

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Cognitive Walkthrough: List of actions



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Cognitive Walkthrough: Questions

4 things needed:

1. a detailed specification of the **system** (or a prototype)
2. a description of the (representative) **task** to perform
3. a complete list of **actions** needed to complete the task (a walkthrough scenario)
4. info about **users**

For each action the evaluators try to answer **four questions**



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Cognitive Walkthrough: Questions

4 questions for each action:

1. **Is the effect of the action the same as the user's goal at that point?**
Will users be trying to produce the right outcome? I.e., does the user understand that this step is needed to reach his goal?



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Cognitive Walkthrough: Questions

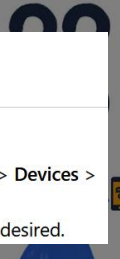
4 questions for each action:

1. **Is the effect of the action the same as the user's goal at that point?**

Turn Bluetooth on or off
Applies to: Windows 10

Here's how to turn Bluetooth on or off in Windows 10:

1. Select the Start button, then select **Settings** > **Devices** > **Bluetooth & other devices**.
2. Select the Bluetooth switch to turn it On or Off as desired.



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Cognitive Walkthrough: Questions

4 questions for each action:

2. **Will users see that the action is available?**
Will users see the correct action they need to perform to produce the outcome? (visibility)

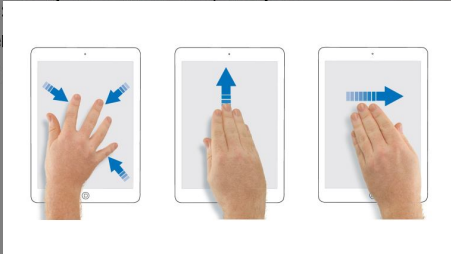


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Cognitive Walkthrough: Questions

4 questions for each action:

2. **Will users see that the action is available?**
Will users need to perform the correct action they need to produce the outcome? (visibility)



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Cognitive Walkthrough: Questions

4 questions for each action:

2. **Will users see that the action is available?**
Will users see the correct action they need to perform to produce the outcome? (visibility)



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Cognitive Walkthrough: Questions

4 questions for each action:

3. **Once users have found the correct action, will they know it is the one they need?**
Will users recognize that this is the action they need to take to achieve the correct outcome? (Or will they select a different action instead?)
 Problems with problems with ambiguous or jargon terms, or with other controls that look like a better choice.



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Cognitive Walkthrough: Questions

4 questions for each action:

4. **After the action is taken, will users understand the feedback they get? Is the feedback appropriate?**
If users performed the correct action, will they be able to tell they made progress toward their intended outcome?



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Cognitive Walkthrough: Questions

4 questions for each action:

4. **After the action is taken, will users understand the feedback they get? Is the feedback appropriate?**
If users performed the correct action, will they be able to tell they made progress toward their intended outcome?



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Cognitive Walkthrough

4 questions for each action

1. Is the **effect** of the action the same as the user's **goal** at that point?
2. Will users **see** that the action is available?
3. Once users have found the correct action, will they know **it is the one** they need?
4. After the action is taken, will users understand the **feedback** they get?



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Cognitive Walkthrough: Focus

The focus of the cognitive walkthrough is on understanding the system's **learnability** for **new** or **infrequent** users.

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Heuristic evaluation

Proposed by **Nielsen** and **Molich**.

Usability criteria (**heuristics**) are identified

Design examined by **experts** to see if these are **violated**

Example heuristics

- system behaviour is predictable
- system behaviour is consistent
- feedback is provided

Debugs **design**

Flexible and **cheap**



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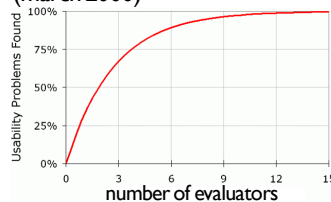
How many evaluators?

Evaluators act **independently** (ideally in sequence..)

According to **Nielsen**, **3-5 evaluators** are sufficient: 5 usually discover about 80% of the usability problems.

<http://www.useit.com/alertbox/20000319.html>

(march 2000)



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Heuristic Evaluation

10 heuristics are provided by Nielsen to aid the evaluators discovering usability problems (supplemented by heuristics specific to the **domain**)

The **severity of each problem** is assessed on the basis of **four factors**:

1. how common is the problem
2. how easy is for the user to overcome it
3. persistence of the problem
4. how seriously will the problem be perceived



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Problem severity

The **severity of each problem** is **rated** on a scale **0-4**

0: it is not a usability problem

1: **cosmetic problem** only; fix only if extra time is available

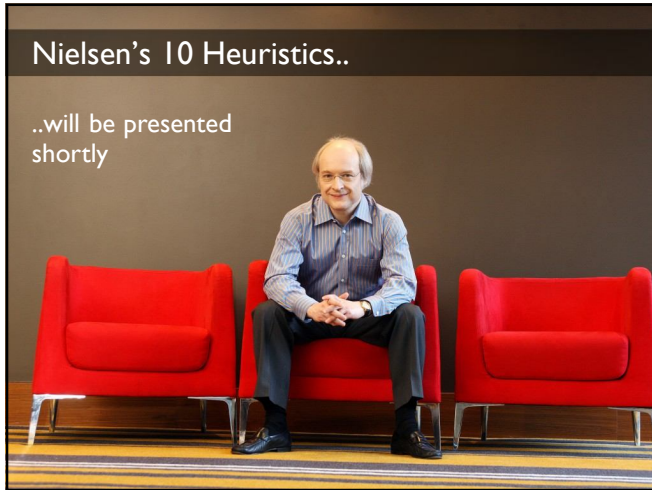
2: **minor usability problem**; fixing has low priority

3: **major usability problem**; important to fix – high priority

4: usability **catastrophe**; imperative to fix



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Using previous studies in evaluation

Results from the literature used to support or refute **parts of design** (i.e. usability of different menu types, choice of icons).

You **save money** by avoiding repeating tests already done by others

Care needed to ensure results are transferable to new design!



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